Chapter 8

Trade

ECON 003
UC Riverside · Summer 2026



Where We Left Off

Chapter 7 asked what one market does when nobody is in charge.
Today we let people trade across two markets — and two countries.

WHAT WE PROVED

A competitive market maximises social surplus. Nobody designs that; the price does it.

WHAT WE ASSUMED

One market, one price, everyone inside the same border.

WHAT WE DROP TODAY

That border. Once two producers can trade, a good can be made wherever it is cheapest.

The gains from trade come from differences in OPPORTUNITY COST, not from differences in skill.

Today's Roadmap

8.1 The production possibilities curve

8.2 Comparative advantage

8.3 Trade between states

8.4 Trade between countries

8.5 Arguments against free trade

TWO PEOPLE

You and Olivia, websites and programs. One of you is faster at everything. Both still gain.

TWO STATES

California and Florida, apricots and bananas. Nobody proposes a tariff at the state line.

TWO COUNTRIES

Denmark and tennis shoes — same arithmetic, but a border can be taxed.

Comparative advantage → specialization → terms of trade → winners and losers. One chain, three scales.

Where the United States Sits in World Trade

Four facts about American trade, before any theory.

MADE ABROAD

~10%

of every dollar Americans spend traces back to an import

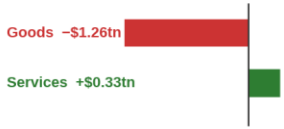


one dollar in ten

THE TRADE DEFICIT

3.0%

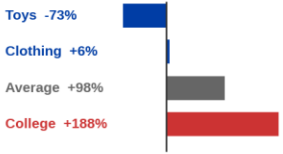
of GDP in 2025 — and the US runs a surplus in services



PRICES SINCE 2000

−73%

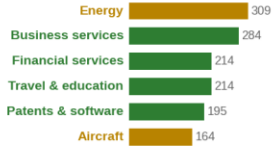
toys. What you can ship got cheaper; what you cannot got dearer



WHAT AMERICA SELLS

$195bn

a year just for the right to use American patents, software, brands



A country sells what it gives up least to produce.

Consumer import share: Federal Reserve Bank of San Francisco, Economic Letters 2019-01 and 2025-12. Trade balance and exports: Census / BEA annual revision, June 2026, 2025 data. Prices: BLS CPI-U, January 2000 to July 2026.

PART 1

The Production Possibilities Curve

First, Trade With Yourself

You have an eight-hour day. You can build websites, or write computer programs, or split the time.

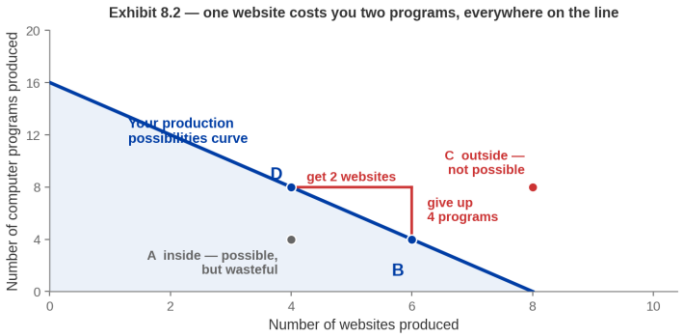
Hours on websites	Websites	Hours on programs	Programs
8	8	0	0
6	6	2	4
4	4	4	8
0	0	8	16



You face a trade off between 1 website and 2 program with each productive hour.

Each website you create has an opportunity cost of 2 programs.

Exhibit 8.2 — The Production Possibilities Curve



WHERE A POINT SITS

ON it (B, D): every hour used. INSIDE (A): possible but wasteful. OUTSIDE (C): not possible today.

THE SLOPE IS THE COST

D to B: give up 4 programs, get 2 websites. So one website costs two programs — everywhere on the line.

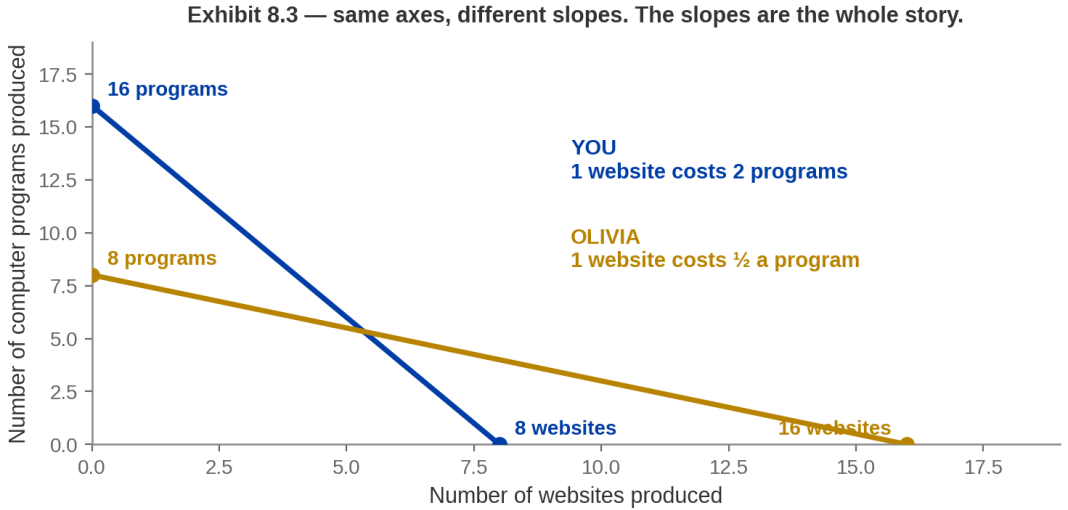
THE DEFINITION

The maximum of one good for a given level of the other. Also called the production possibilities FRONTIER.

Same shape as Chapter 5's budget constraint — but this line is what you can PRODUCE, not what you can afford.

Exhibit 8.3 — Two Production Possibilities Curves

Olivia has the same eight hours and the opposite strengths.



The slope of a curve IS that person's opportunity cost. Different slopes, different costs.

Calculating Opportunity Cost

OPPORTUNITY COST — the units of the other good you give up to produce one unit of this one.

	One website costs	One program costs
You	2 programs	½ a website
Olivia	½ a program	2 websites

Websites are cheap for Olivia. Programs are cheap for you. Nobody is cheapest at both.

Different people have different specialties — this is where the DIVISION OF LABOUR arises.

PART 2

The Basis for Trade: Comparative Advantage

Comparative Advantage

COMPARATIVE ADVANTAGE — the ability of one economic agent to produce at a lower opportunity cost than another.

YOU

Lower cost of producing PROGRAMS: half a website each, against Olivia's two.

OLIVIA

Lower cost of producing WEBSITES: half a program each, against your two.

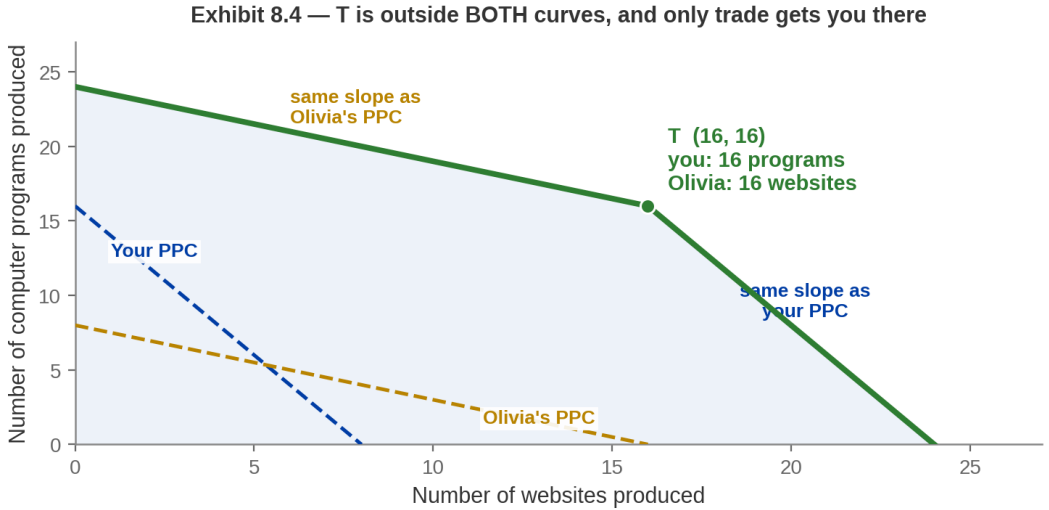
SO SPECIALIZE

Each of you does only the good you are cheapest at, and then you trade. More gets produced out of the same sixteen hours.

The person with the lowest opportunity cost should be the one who produces.

Exhibit 8.4 — The Gains from Specialization

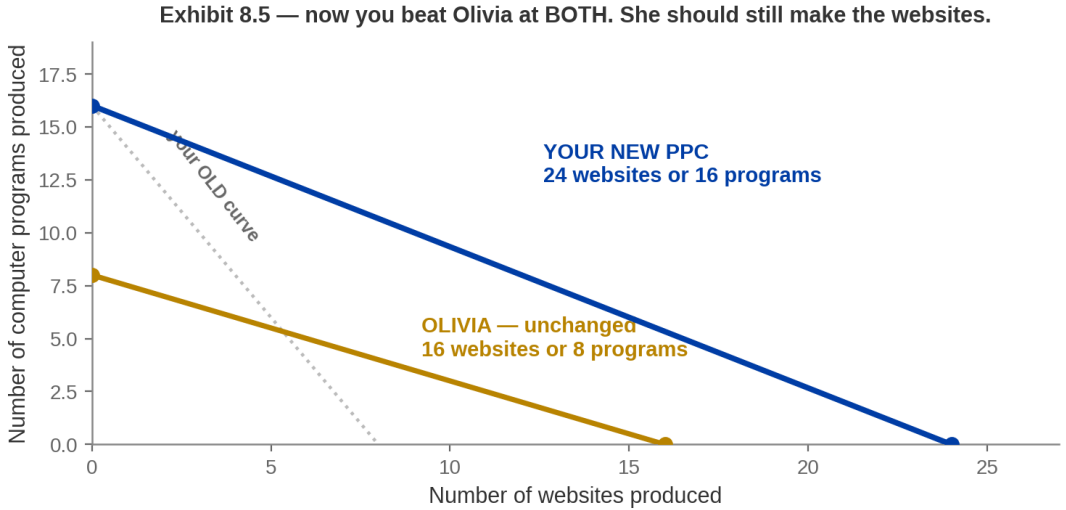
Add the two curves together. You make only programs, Olivia makes only websites.



Point T lies outside BOTH curves: specialization lets you consume beyond your own production possibilities.

Exhibit 8.5 — What If You Were Better at Both?

You take a course in web design. You can now make 24 websites a day, or 16 programs.





You now out-produce Olivia at BOTH goods. Should you just do everything yourself?

Absolute Advantage — and Why It Does Not Decide

ABSOLUTE ADVANTAGE — the ability to produce MORE output than another agent with the same resources.

	One website costs	One program costs
You (after the course)	⅔ of a program (0.67)	1½ websites (1.50)
Olivia	½ a program (0.50)	2 websites (2.00)

0.50 beats 0.67, so websites are STILL Olivia's. 1.50 beats 2.00, so programs are still yours.

You have both absolute advantages and neither of you has both comparative advantages unless the productivity ratios are the same for the two of you.

Would You Take This Trade?

You can always "trade with yourself" at your own opportunity cost. Any offer has to beat that.

1 WEBSITE FOR 1 PROGRAM?

NO. Give up a program yourself and you get 1½ websites. Olivia's offer is worse than doing it alone.

2 WEBSITES FOR 1 PROGRAM?

YES. Two beats the 1½ you could get on your own. You are better off trading.

3 WEBSITES FOR 1 PROGRAM?

Olivia says no. She can get 2 websites per program by herself, so 3 is above her limit.

Your minimum is 1½ websites. Her maximum is 2 websites. Any price strictly between them works.

Terms of Trade

TERMS OF TRADE — the "price" of one good in terms of the other; the exchange rate between goods.



THE MODEL FIXES THE RANGE

Between the two opportunity costs — here, 1½ to 2 websites for a program. Outside that range one side is better off not trading at all, so no deal happens.

BUT NOT THE PRICE INSIDE IT

Where it lands depends on bargaining, or on a market. Near 1½ Olivia captures most of the gain; near 2 you do. Both gain either way — but not equally.

Trade creates surplus; the terms of trade split it.

PART 3

Trade Between States

Exports and Imports

EXPORT — produced here, shipped to another state or country.
IMPORT — produced in another state or country, sold here.



NOTICE WHAT IS MISSING

Neither definition mentions a national border. The book defines both for STATES as well as countries, on purpose.

WHAT GIVES A STATE ITS EDGE

Natural resources, proximity to rivers, ports and transport hubs, and concentrations of expertise.

AND THE CONSEQUENCE

States specialize hard and trade the results. No one regards that as a problem, and no one proposes a tariff at the state line.

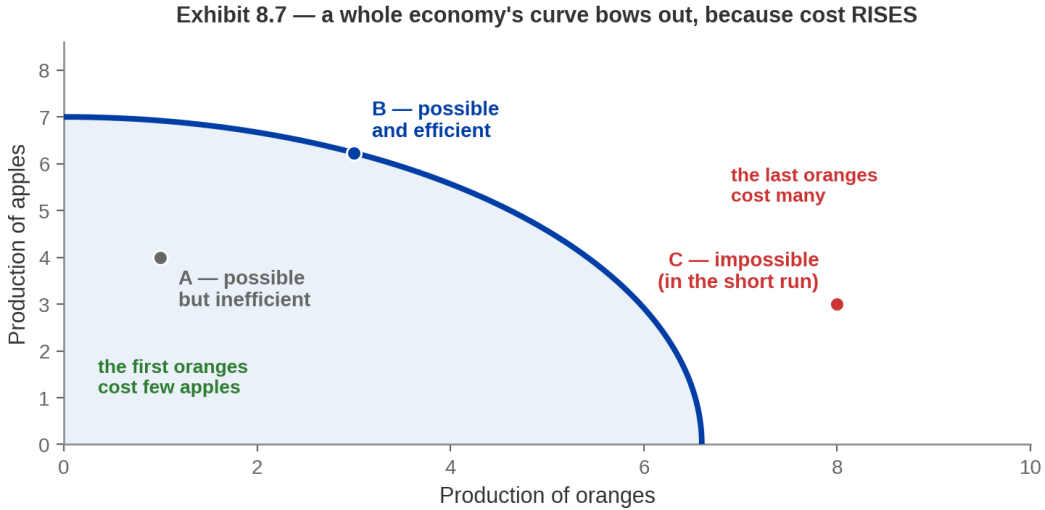






Exhibit 8.7 — A Whole Economy's Curve Bends

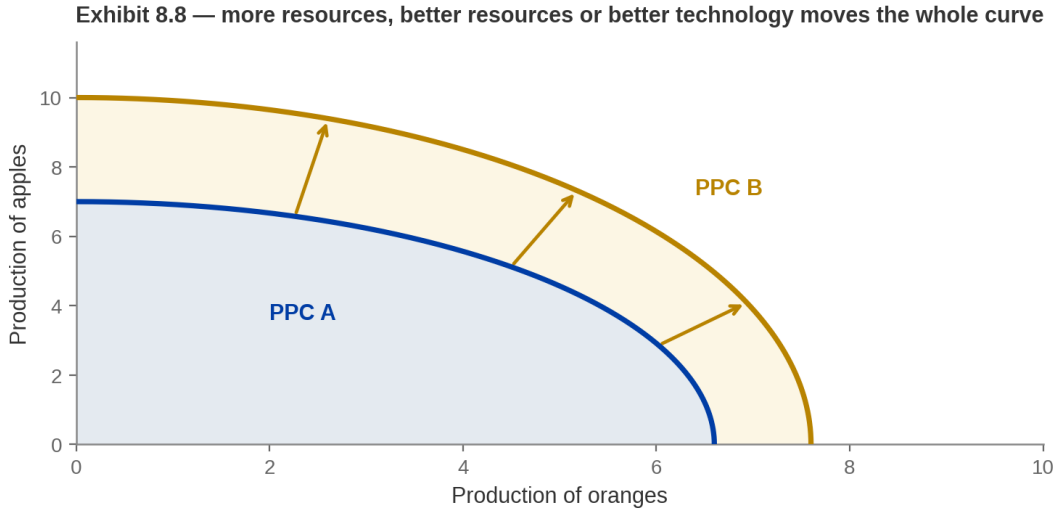
INCREASING OPPORTUNITY COST — each extra unit costs more of the other than the last.



Resources are not equally suited to every use, so an economy's curve bows outward.

Exhibit 8.8 — How Improved Technology Shifts the Curve

Point C was impossible. What would it take to make it possible?



ECONOMIC GROWTH is the production possibilities curve shifting outward.

Who Should Produce What? California and Florida

The same opportunity-cost table, for two states instead of two people.





	One apricot costs	One banana costs
California	⅕ of a banana	5 apricots
Florida	8 bananas	⅛ of an apricot

Apricots are cheap in California (⅕ against 8). Bananas are cheap in Florida (⅛ against 5). So the terms of trade for a banana lie between ⅛ and 5 apricots.

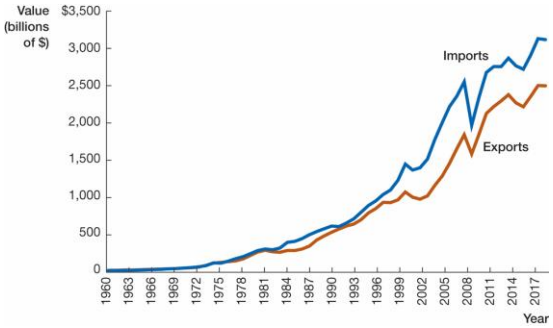
Comparative advantage does not depend on the size of the trading unit: person, state or country.

PART 4

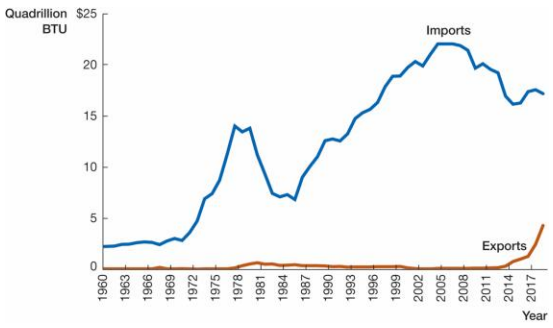
Trade Between Countries

What the United States Actually Trades

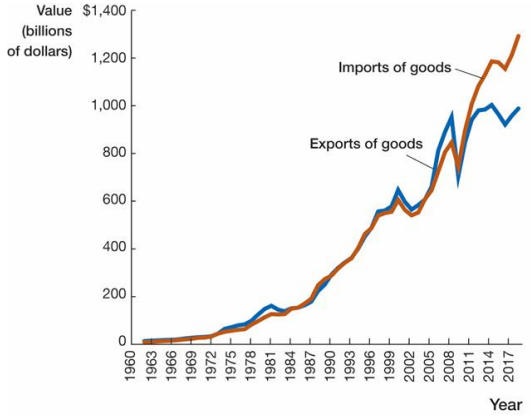
TRADE DEFICIT — imports exceed exports. **TRADE SURPLUS — exports exceed imports.**



The aggregate



Crude oil



Manufactured goods

"Net importer" is a statement about a total. It says almost nothing about any particular good.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Trade Deficits – Let an Expert Explain



Where Do World Prices Come From?

WORLD PRICE — where world supply of a good meets world demand.

WHAT A COUNTRY STARTS WITH



Natural resources, the capital it has already built, and its technology.

WHO DOES THE WORK



Education, work habits and experience of the labor force — and how abundant labor is against physical capital.

AND THE ACCIDENTS



Climate. Denmark does not grow bananas, and no amount of policy will change that.

Differences in resources, capital and technology are what make one country's costs differ from another's.

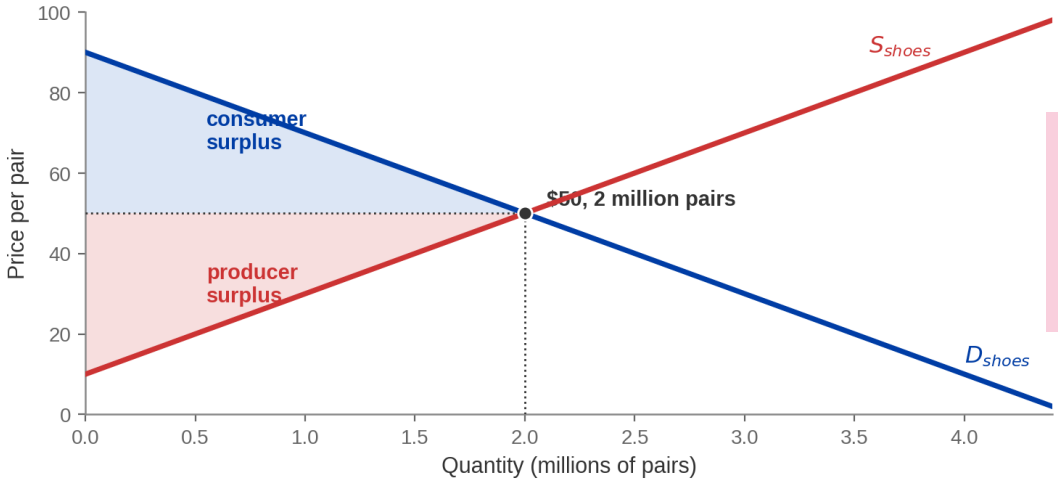
Start With the Border Closed - Autarky

The market for tennis shoes in Denmark, with no imports and no exports.





Exhibit 8.12 — Denmark's tennis-shoe market with the border closed

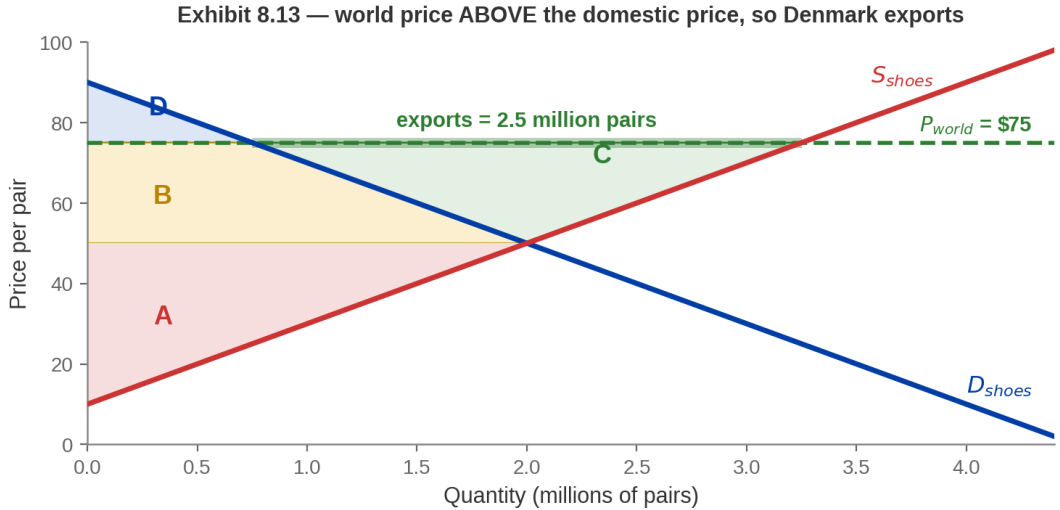




Denmark is a small country that has little influence over the world price.

Winners and Losers in an Exporting Nation

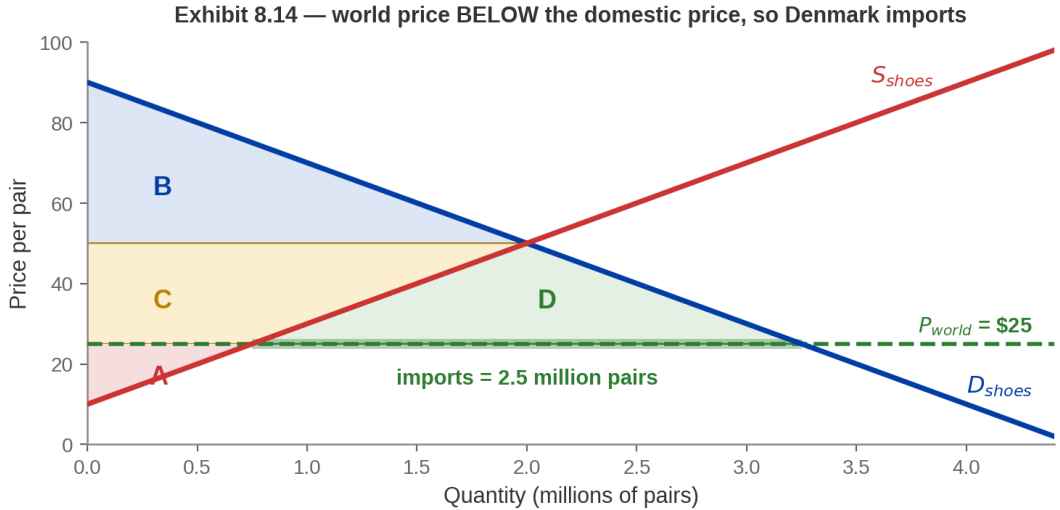
Suppose the world price is $75. No Danish producer will sell at home for $50.



Producers gain B + C, consumers lose B, and the country nets area C — 31¼ million dollars.

Winners and Losers in an Importing Nation

Now the world price is $25 — BELOW the domestic $50. Denmark imports.



Consumers gain C + D, producers lose C, and the country nets area D — again 31¼ million dollars.

The rule is: a world price BELOW the domestic price means you import. Some printed versions of this exhibit state it the wrong way round.

Either Direction, the Country Gains

Exporting: producers gain more than consumers lose. Importing: consumers gain more than producers lose.

THE RESULT

In both cases Denmark is better off overall. The specific winners and losers swap; the total goes up either way.

THE CATCH

There is no version of opening a border where everyone at home gains. Somebody was benefiting from it being shut.

THE BOOK'S ANSWER

Winners COULD be required to compensate losers — and still come out ahead. Whether they are is a political choice.

"The country gains" is a claim about a total. It never was a claim that each person gains.

PART 5

Arguments Against Free Trade

Not Everybody Is Convinced



Every result so far said trade makes a country better off.
So why do people protest against it?

Because “better off” has been a statement about a TOTAL, all hour. The diagrams have been showing you the losers in red since Danish shoemakers when the price fell, Danish shoe buyers when it rose.

Those people are real, they know exactly who they are, and the compensation the model says is affordable is rarely actually paid.

Image: Acemoglu, Laibson & List, Microeconomics 3e

Arguments Against Free Trade

Every one of them accepts the diagram and argues that something outside it matters more.

- National security
- Culture and globalization
- Environment and resources
- Infant industries
- Local wages and jobs

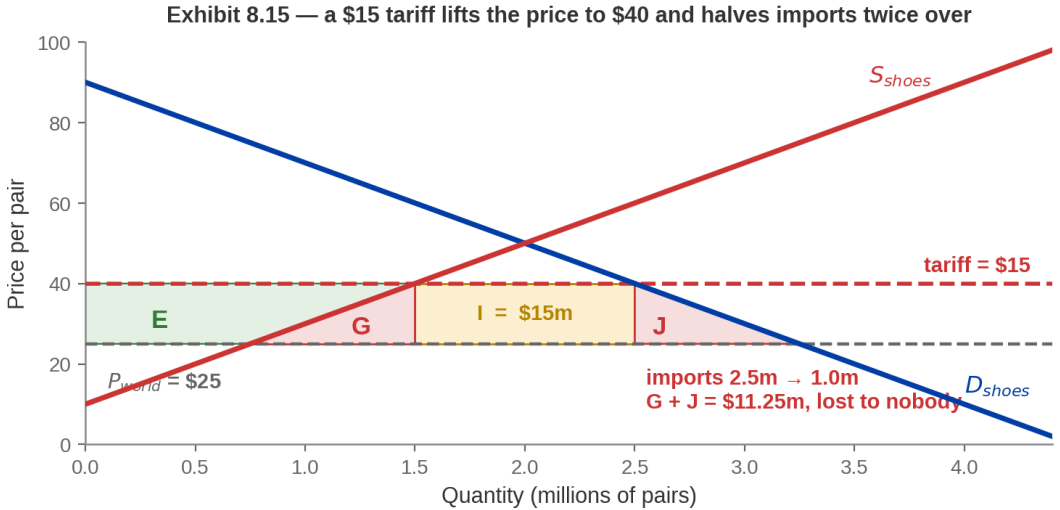
PROTECTIONISM — the view that governments should control trade because of the harmful effects of free trade.
Its most popular instrument is the TARIFF.



The test for any of them: is a tariff the cheapest way to buy the thing you actually want?

The Effect of a Tariff

A TARIFF is a tax added to the price of an imported product.



A $15 tariff lifts the Danish price from $25 to $40, and imports fall from 2.5 million pairs to 1 million.

Who Gets What When a Tariff Goes On

E → PRODUCERS

A higher price on a larger domestic output. Danish shoemakers gain, and they are easy to identify.

I → GOVERNMENT

$15 a pair on the 1 million pairs still imported. Exactly $15 million.

G → NOBODY

The production distortion. Pairs now made in Denmark above the $25 the world would have charged.

J → NOBODY

The consumption distortion. Pairs Danes valued above $25 that are simply never bought.

E and I are TRANSFERS between Danes. G + J = $11.25 million is not transferred anywhere — it stops existing.

G + J is DEADWEIGHT LOSS — surplus that no one receives, exactly as under a price control.

Will Free Trade Cause You to Lose Your Job?

Trade destroys jobs in the importing industry and creates them in the exporting one.

SOME WORKERS DO LOSE

Fewer shoes are made in Denmark once it imports them. That can depress wages or put cobblers out of work, and it is not hypothetical.

BUT THERE IS NO BROAD HARM

There is no systematic evidence that opening up to trade harms workers as a whole. The wider gains offset much of the loss, so the net effect on jobs can be quite small.

THE CATCH IS COMPENSATION



In theory the winners can more than compensate the losers. In practice, complete compensation is difficult — and rarely required.

Compensation is affordable in the model and rare in practice.

Chapter 8: The Full Picture

	What it says	Key terms
The curve	All you can make with the hours you have. Its slope is your cost.	production possibilities curve
Who does what	Do what costs YOU least — not what you are best at.	comparative advantage, absolute advantage
The gains	Specialise, then trade at a price between the two costs.	specialization, terms of trade
Bigger units	An economy holds many costs, so its curve bows outward.	increasing opportunity cost
Countries	Compare the world price with your own. Below it you import.	world price, exports, imports
Protection	A transfer to producers and government, plus a loss nobody gets.	tariff, protectionism

One idea, proved three times: people, states and countries all gain by doing what they give up least to do.

KEY TAKEAWAYS

Chapter 8: Trade

- The production possibilities curve shows what you can make from the resources you have. Its slope is an opportunity cost.
- The basis for trade is comparative advantage — lowest opportunity cost — and never absolute advantage.
- Specialise and trade at a price between the two opportunity costs, and both sides end up beyond their own curve.
- There are winners and losers in every trading state and country, and which group loses depends only on which way the goods move.
- The winners can more than compensate the losers. Whether they do is a political question, not an economic one.